

Reciprocal Transfer Effects of Suprasegmental Features of First and Second Language examined through Serbo-Croatian Pitch Accent in Heritage Speakers

Erdin Mujezinovic

Heinrich Heine University Düsseldorf: Department of General Linguistics

Objectives

- 1. Attesting, whether suprasegmental features are more likely to transfer from dominant to heritage language, or vice-versa.
- 2. Collect supporting evidence for **age of acquisition effects**.
- 3. Highlighting the role of prosody transfer in bilingual speakers language production.

Introduction

- Serbo-Croatian is an Indo-European, South Slavic language spoken in Serbia, Croatia, Montenegro, Bosnia/Herzegovina and Kosovo (Kordić, 2008).
- The Serbo-Croatian prosodic system consists of falling **FA** and rising **RA** contour tones; pitch is utilized contrastively at the lexical stratum (Lehiste and Ivić, 1986; Inkelas and Zec, 1988; Petrović, 2006).
- FA: /â/, /â:/
Examples: [sân] ‘dream’, [mrâ:k] ‘darkness’, [mât̪ka] ‘cat’, [zâ:stava] ‘flag’
- RA: /ǎ/, /ǎ:/
Examples: [mǎgla] ‘fog’, [xǎ:la] ‘gratitude’, [budǎla] ‘idiot’, [muǰkǎ:rats] ‘man’

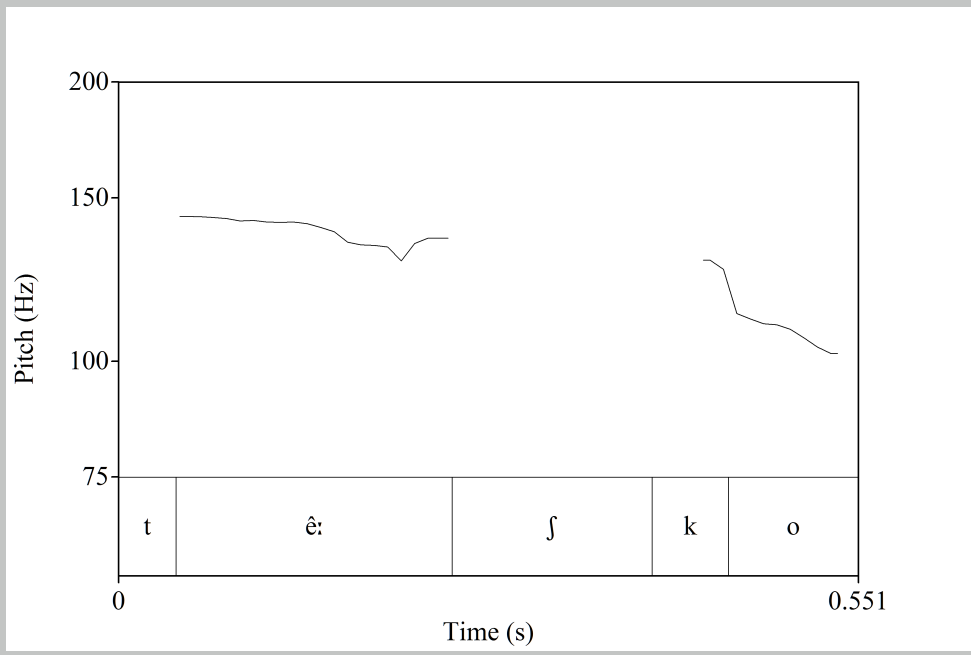


Figure 1: Pitch track (F₀) for a Serbo-Croatian word with FA

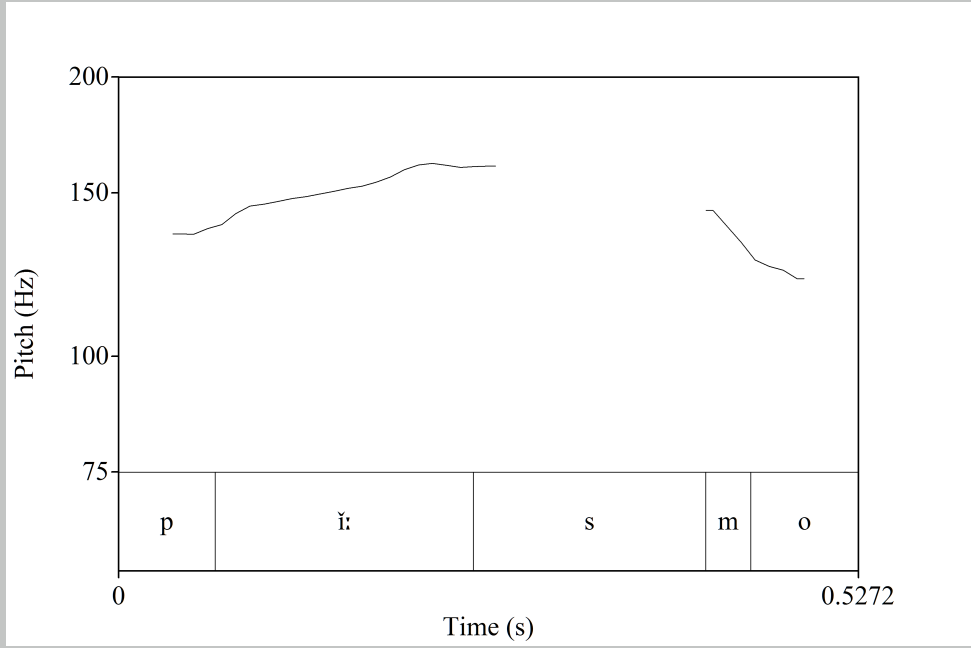


Figure 2: Pitch track (F₀) for a Serbo-Croatian word with RA

Language Differences

- Serbo-Croatian = *lexical pitch accent language* (Lehiste and Ivić, 1986; Zsiga, 2013)
- German = *lexical stress language* with free, but quantity sensitive stress distribution (Domahs, Plag and Carrol, 2014)
 - While pitch contour has no distinguishing function, lexical stress is partly realized through an increase in relative pitch, with duration being a stronger correlate (Kleber and Klippahhn, 2006).
- Work on transfer from pitch-accent to stress is fairly limited.
- German heritage speakers of Serbo-Croatian were compared to monolingual speakers in both languages.
- Heritage speakers’ L1 (Serbo-Croatian) suffers from language attrition; their L2 (German) is used most of the time.

Materials

- 40 Serbo-Croatian words, collected through the *SRPSKIJE online-dictionary* (Hälg, Adžaić and Novković, 2007); 40 German words.
- Serbo-Croatian: 20 with RA, 20 with FA; balanced in vowel quality and quantity for the tonic syllable.
- German: 20 monosyllabic, 20 multisyllabic words with stress at the penultima; vowel types for the accented syllables were balanced.
 - This distinction was chosen in order to replicate the context, where in Serbo-Croatian only one type of contour tone can occur.
 - FA -> only in monosyllabic words and in syllable-initial position
 - RA -> only in multisyllabic words (initial or medial position)

Methods

- Reading experiments for each respective language
- Items were embedded in a carrier sentence
 - *Na tabli riječ ITEM je napisana* (Serbo-Croatian)
 - *Ich habe ITEM gesagt* (German)
- F₀-slopes for tonic syllables (Serbo-Croatian) and accented syllables (German) were calculated with *Praat* (Boersma and Weenink, 2018).

Participants

- Participants language proficiency and background was attested by means of self assessment.
- 8 Serbo-Croatian heritage speakers **HS**, living in Germany (age = Ø22).
 - At least a week in between German and Serbo-Croatian recordings.
- 8 Serbo-Croatian speakers **SC**, raised in monolingual environment (age = Ø47).
- 8 German speakers **GER**, raised in monolingual environment (age = Ø35).

Results: Serbo-Croatian

- F₀-slope = (log2(maximum F₀)/(minimum F₀))*12
 - Converted from Hertz to Semitones
 - Dependent variable
- *Linear mixed effects models* (Kuznetsova, Brockhoff and Christensen, 2017) were calculated, with *items* and *participants* as random effects.
- Results for tonic syllables:
 - *Group* was significant ($\chi^2(1) = 123.69$, $p < 2.2e-16$).
 - *Tone type* was significant ($\chi^2(1) = 10.265$, $p = 0.001356$).
 - *Interaction between group and tone type* was **non significant** ($\chi^2(1) = 1.7174$, $p = 0.19$).

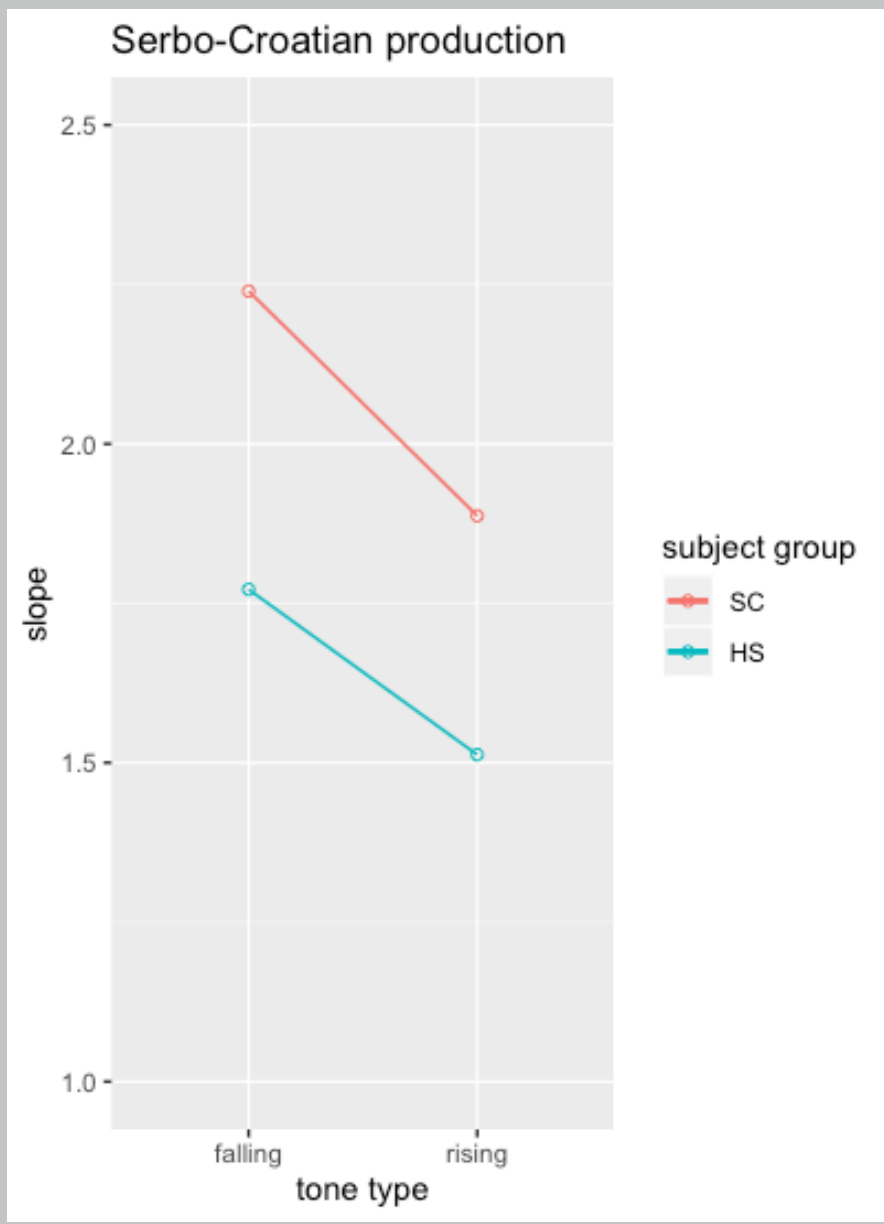


Figure 3: Mean F₀-slope in Serbo-Croatian experiment

Results: German

- Results for accented syllables:
 - *Group* was significant ($\chi^2(1) = 22.068$, $p < 0.01$).
 - *Syllable pattern* was non significant ($\chi^2(1) = 0.175$, $p = 0.6757$).
 - *Interaction between group and syllable pattern* was **significant** ($\chi^2(1) = 3.868$, $p = 0.04921$).
- Only the HS group differed in F₀-slope between monosyllabic and multisyllabic German words.

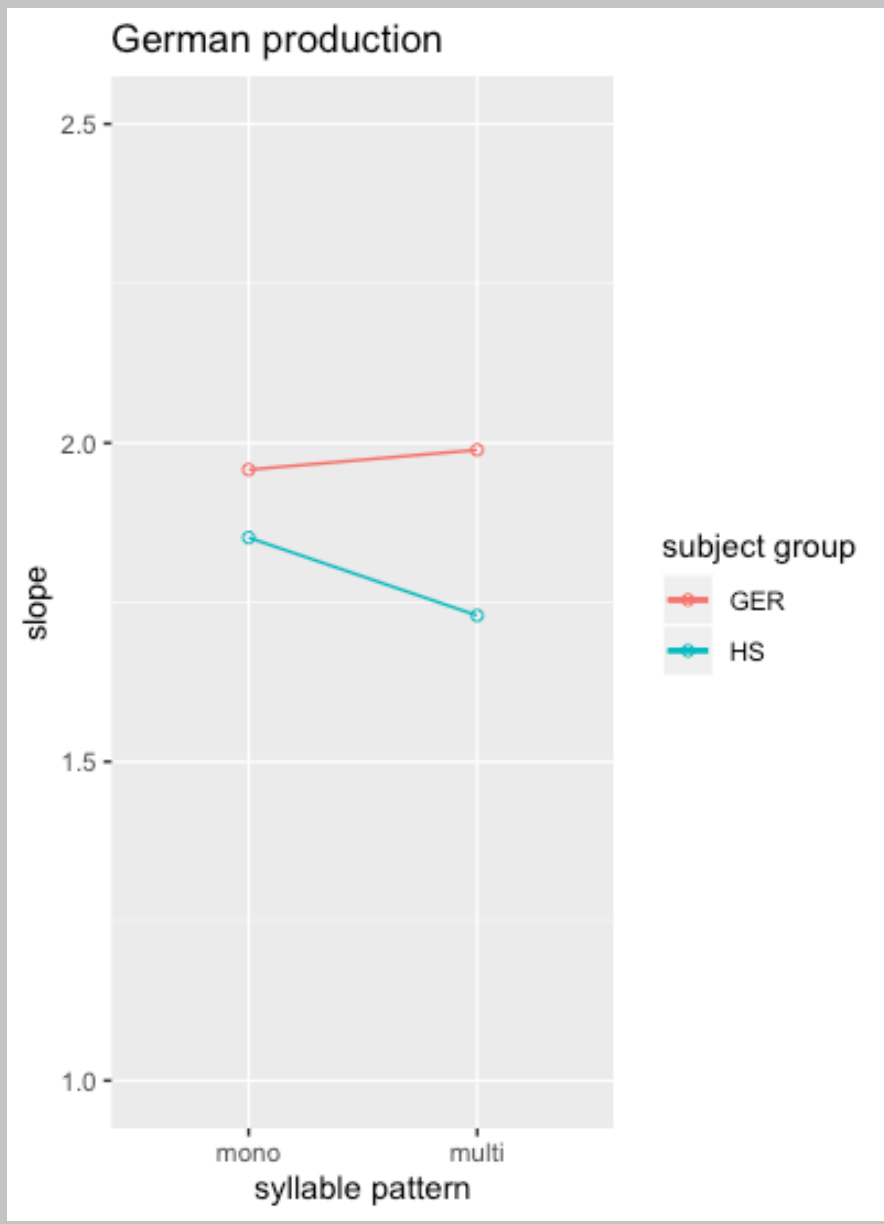


Figure 4: Mean F₀-slope in German experiment

Conclusion

- Even if heritage speakers are more proficient in their L2, their L2 prosody is still effected by the L1.
- F₀-pattern reflects their realization of falling and rising tones in SC (although to a lesser degree).
- The *Perceptual Assimilation Model* (Best et al., 1995) might account for prosody transfer as well.
- Rising tones are more marked than falling tones (Zec and Zsiga, 2010) and thus more likely to transfer.
- Results do not clearly indicate, if falling tones are transferred as well.

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